

Data Processing Unit Board

Testing Guide

Version	Update Date	Author	Memo
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1. Hardware Preparation

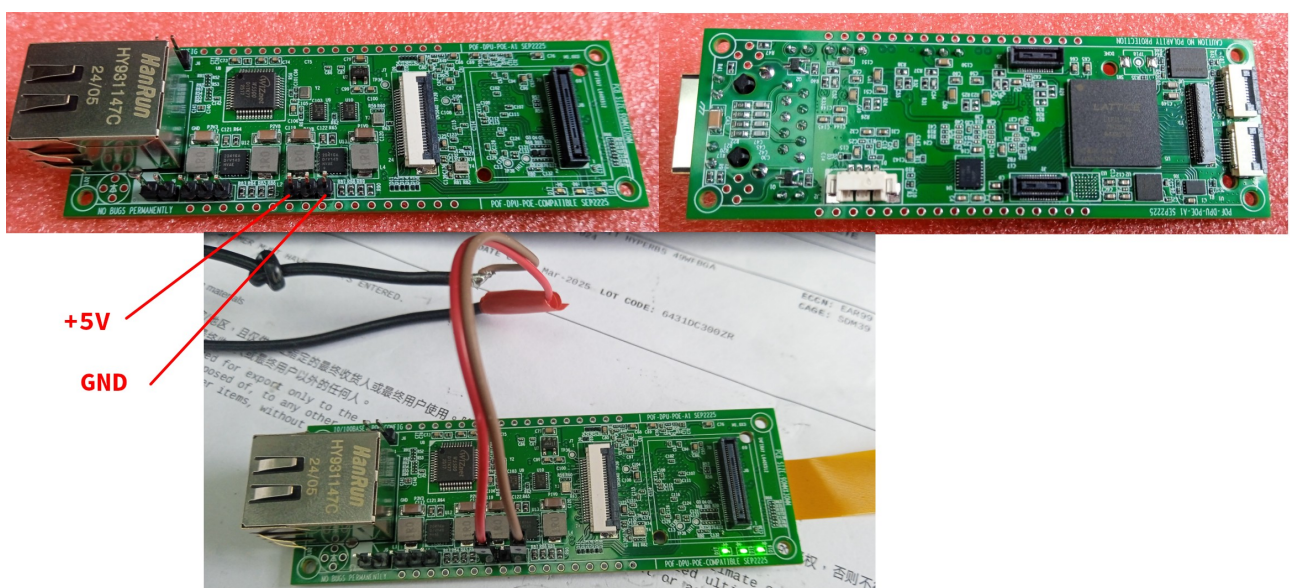
This guide demonstrates how to test the DPU board in comprehensive. Although this board is designed to get power supply from PoE (Power Over Ethernet), it also can be powered from lab power supply. Before testing, please make sure all components below are ready.

No.	Components Name	Memo
1	DuPont Wire	Connect DPU Board to Lab Power Supply
2	DPU Board	One piece of DPU Board
3	Ultrasonic Sensor PreAMP Board	One piece of Ultrasonic Sensor PreAMP Board
4	Oscilloscope	Check waveform.
5	Oscilloscope Probe	Probe x10
6	Digital Multi-meter	Flue Multi-meter
7	Lab DC Power Supply	5V Voltage Source
8	PoE Injector	Supports both Alternative A and Alternative B

2. Hardware Connection

2.1 Connect DPU to DC Power Supply

Connect DPU board to Lab DC Power Supply with DuPont wires. Ensure the correct pinout before connecting. The left pin is 5V, the middle one is floating, the right is GND. Warning, no polarity protection circuit on board due to PCB size.



2.2 Set Over Current Threshold and Power On

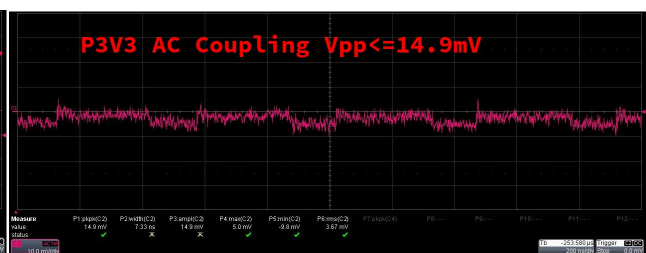
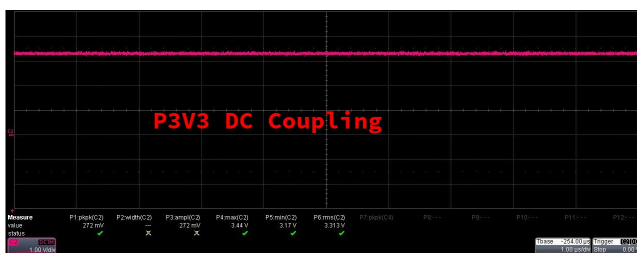
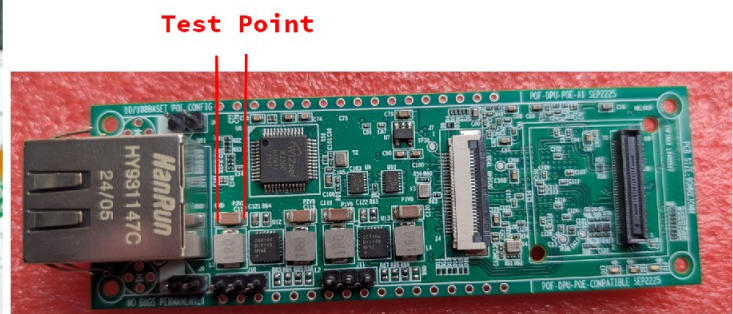
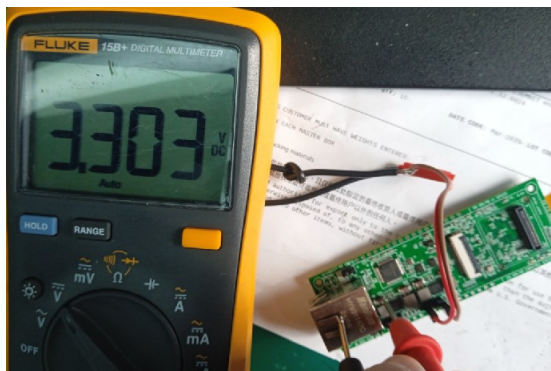
At the 1st time, set a lower over current threshold in order to cut off power supply immediately in case of short-circuit. Experience shows 200mA is appropriate. The current consumption accelerates as FPGA logic expands. If over current occurs, troubleshoot the hardware carefully.



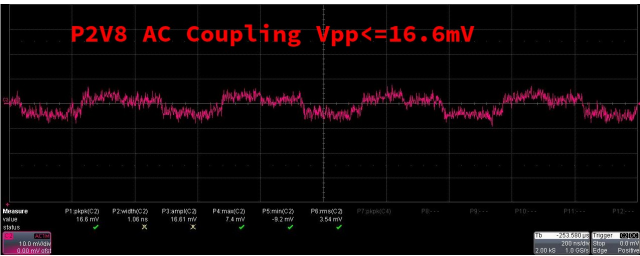
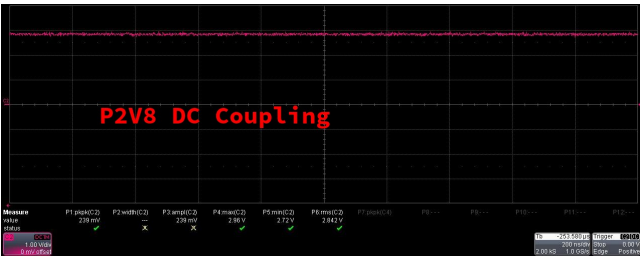
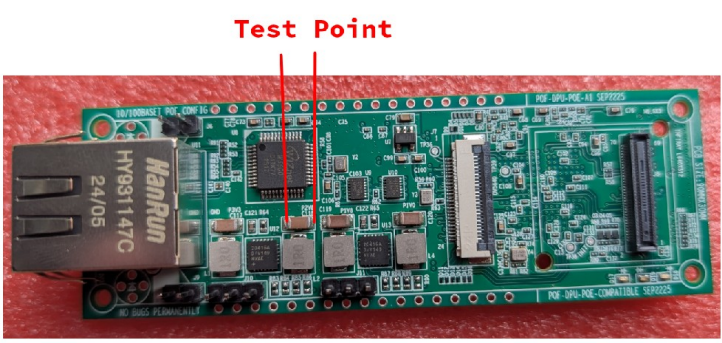
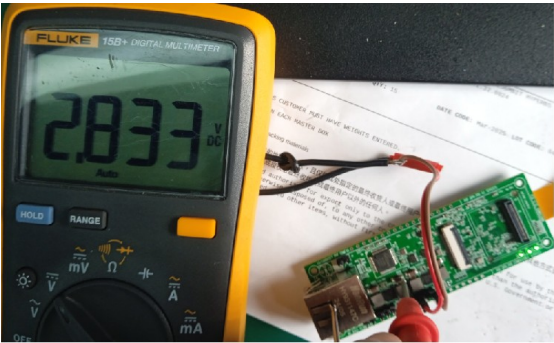
3. Power Rails Ripple Voltage Test

3.1 P3V3 Ripple Voltage

Measure P3V3 power rail ripple voltage at the point described below. Avoid long ground loops when using an oscilloscope probe. $V_{pp} \leq 22.74\text{mV}$



3.2 P2V8 Ripple Voltage

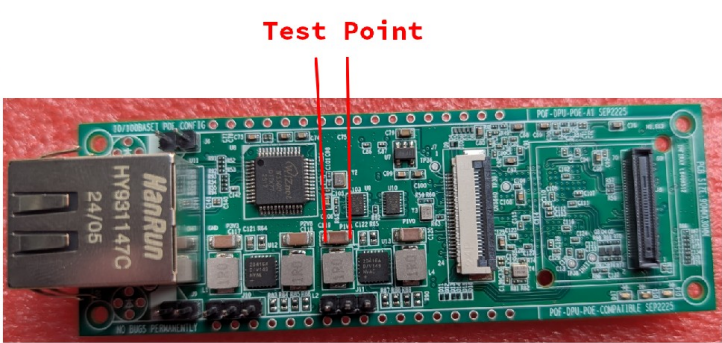
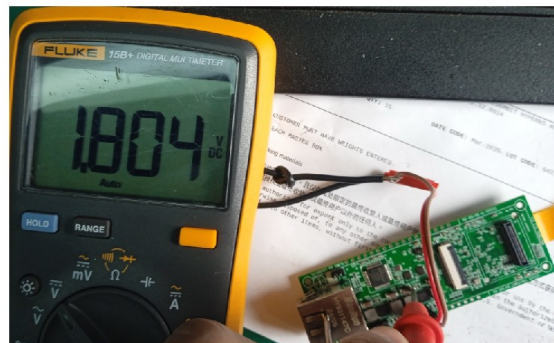


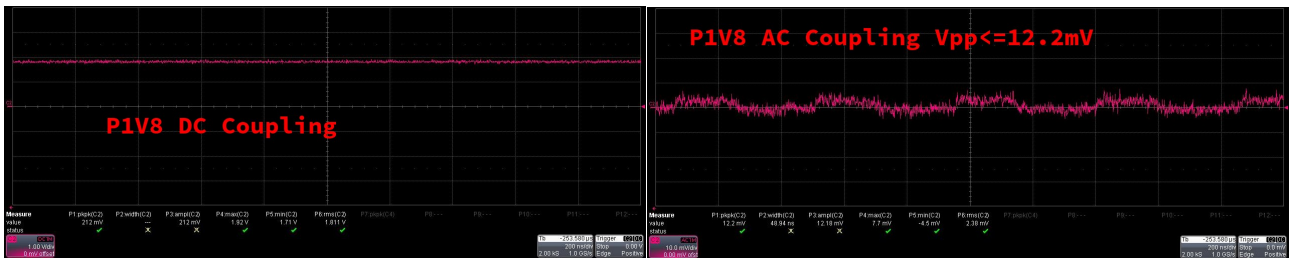
3.3 P1V8 Ripple Voltage

P1V8 supplies I/O drive circuit and auxiliary parts. It's voltage tolerance is between 1.746 and 1.89V. Thus, $1.8\text{V} - 1.746\text{V} = 54\text{mV}$. $1.89\text{V} - 1.8\text{V} = 90\text{mV}$. Ensure the voltage ripple is less than 54mV.

Table 3.2. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Typ.	Max	Unit
V_{CC} , V_{CCCLK}	Core Supply Voltage	$V_{CC} = 1.0$	0.95	1.00	1.05	V
V_{CCAUX}	Auxiliary Supply Voltage	Bank 0, Bank 1, Bank 2, Bank 6, Bank 7	1.746	1.80	1.89	V
$V_{CCAUXH3/4/5}$	Auxiliary Supply Voltage	Bank 3, Bank 4, Bank 5	1.746	1.80	1.89	V
V_{CCAUXA}	Auxiliary Supply Voltage for core logic	—	1.746	1.80	1.89	V
V_{CCIO}	I/O Driver Supply Voltage	$V_{CCIO} = 1.8\text{ V}$, All Banks	1.71	1.80	1.89	V



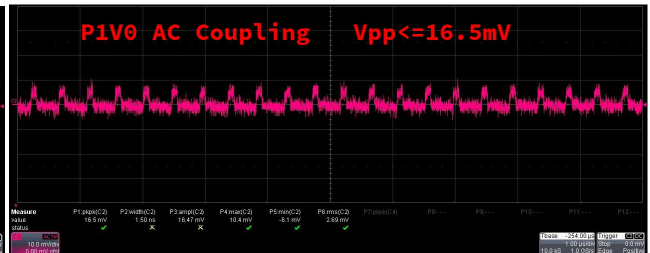
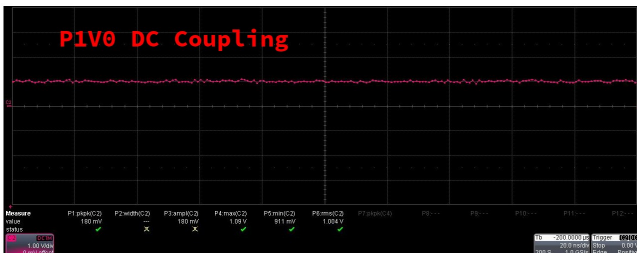
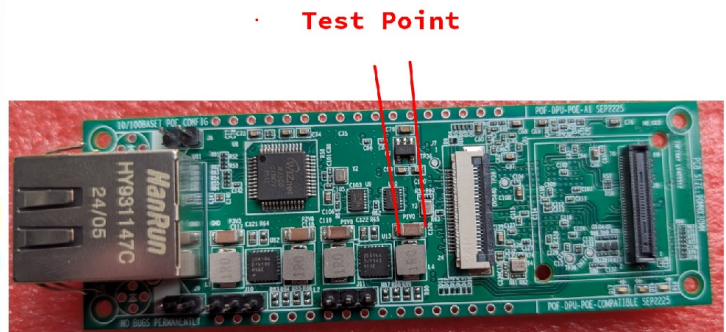
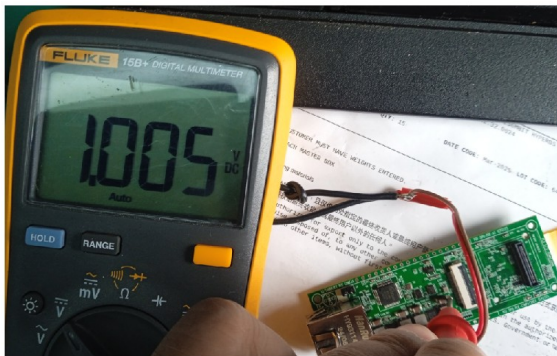


3.4 P1V0 Ripple Voltage

FPGA core works at a low voltage of 1.0V. Its official manual shows the voltage tolerance is between 0.95V~1.05V, ensure the ripple voltage is less than 50 millivolt.

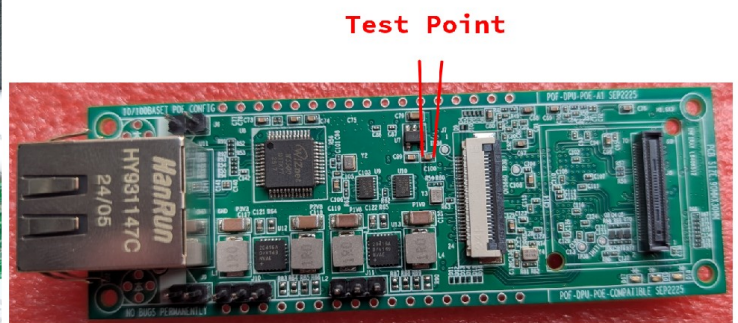
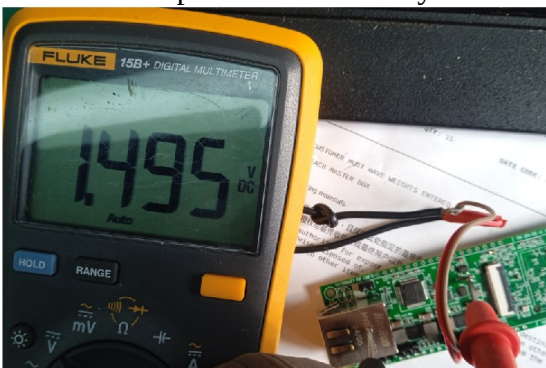
Table 3.2. Recommended Operating Conditions

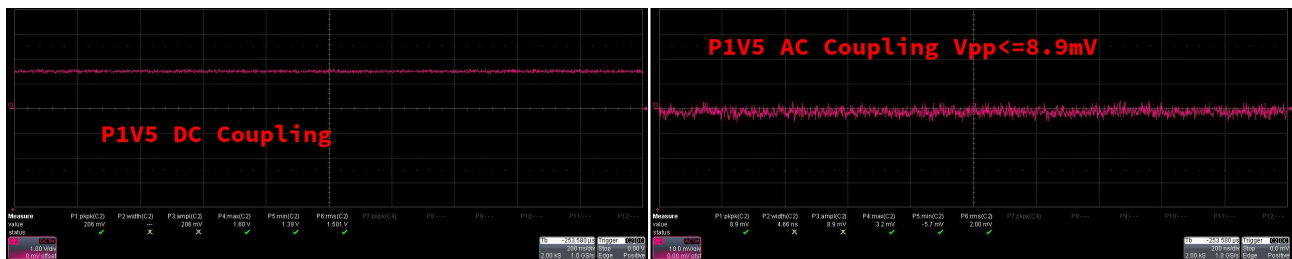
Symbol	Parameter	Conditions	Min	Typ.	Max	Unit
V_{CC}, V_{CCECLK}	Core Supply Voltage	$V_{CC} = 1.0$	0.95	1.00	1.05	V



3.5 P1V5 Ripple Voltage

P1V5 power rail is mainly used for OV5640, so it's placed near the DVP connector.

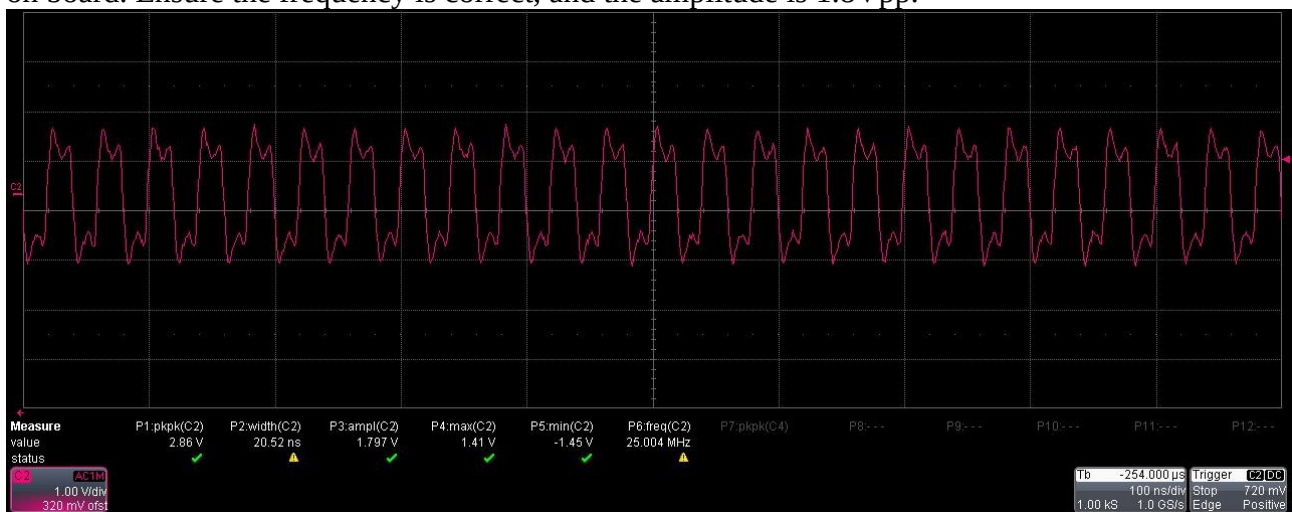




4. Oscillator Frequency Test

4.1 FPGA External Clock – 25.0MHz

Although LIFCL-40 has built-in clocks, an external oscillator of 25.0MHz is also soldered on board. Ensure the frequency is correct, and the amplitude is 1.8Vpp.



4.2 OV5640 Clock – 24.0MHz

Visible Light Image Sensor OV5640 uses an external oscillator of 24.0MHz, ensure the frequency is correct, and the amplitude is 1.8Vpp.



4.3 W5500 Clock – 25.0MHz

W5500 Ethernet Chip uses an external crystal of 25MHz, ensure the frequency is correct, and the amplitude is 1.8Vpp.



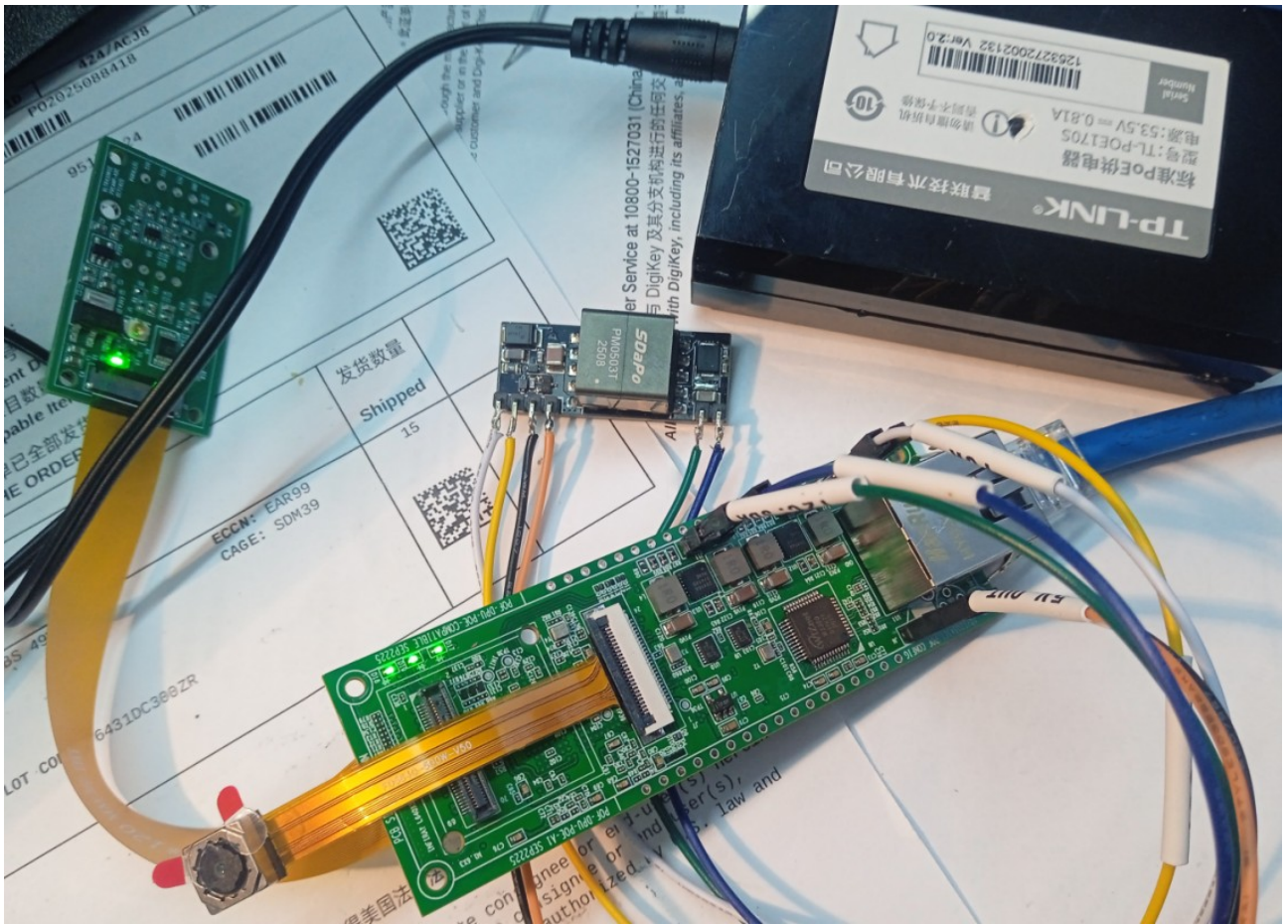
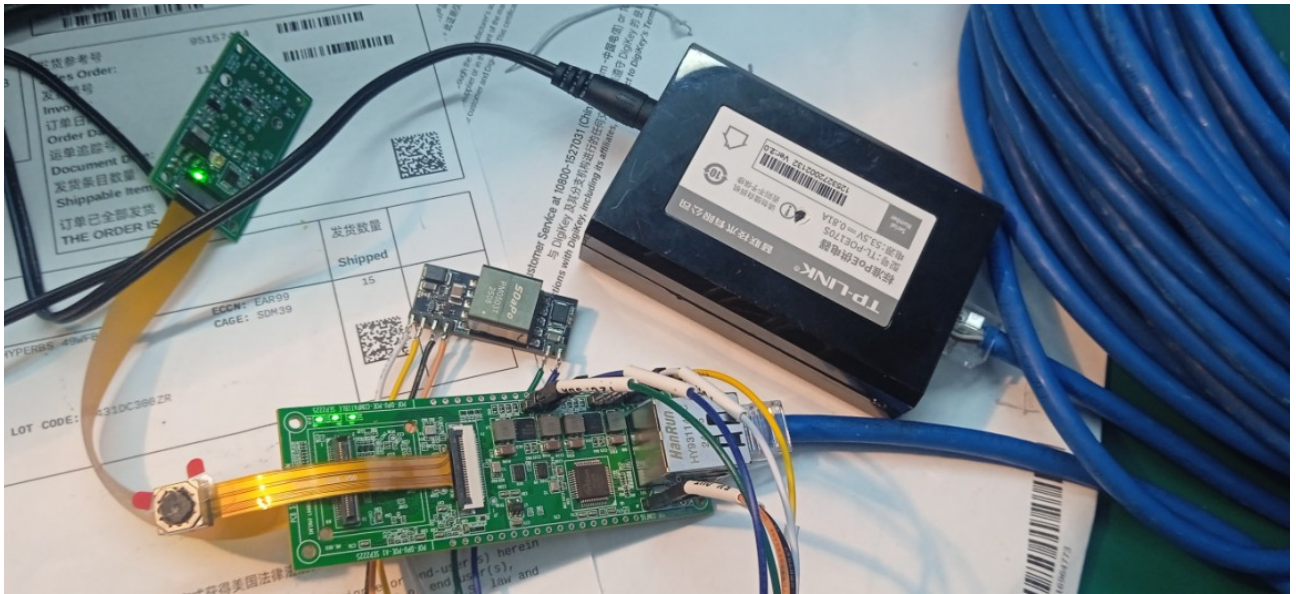
5. Network Cable Connection Test

At default, the two LEDs will be turned on and flash when plug network cable. But make sure the ETH_RST(Pin 37) is pulled to high to set W5500 in normal mode, pulled to low will force it enters RST mode. So please drive ETH_RST to HIGH in FPGA logic.

RJ45 has two built-in LEDs. The Linked LED is on when cable connects, the ACT LED flashes when W5500 receives network data frames even without configure data written in.

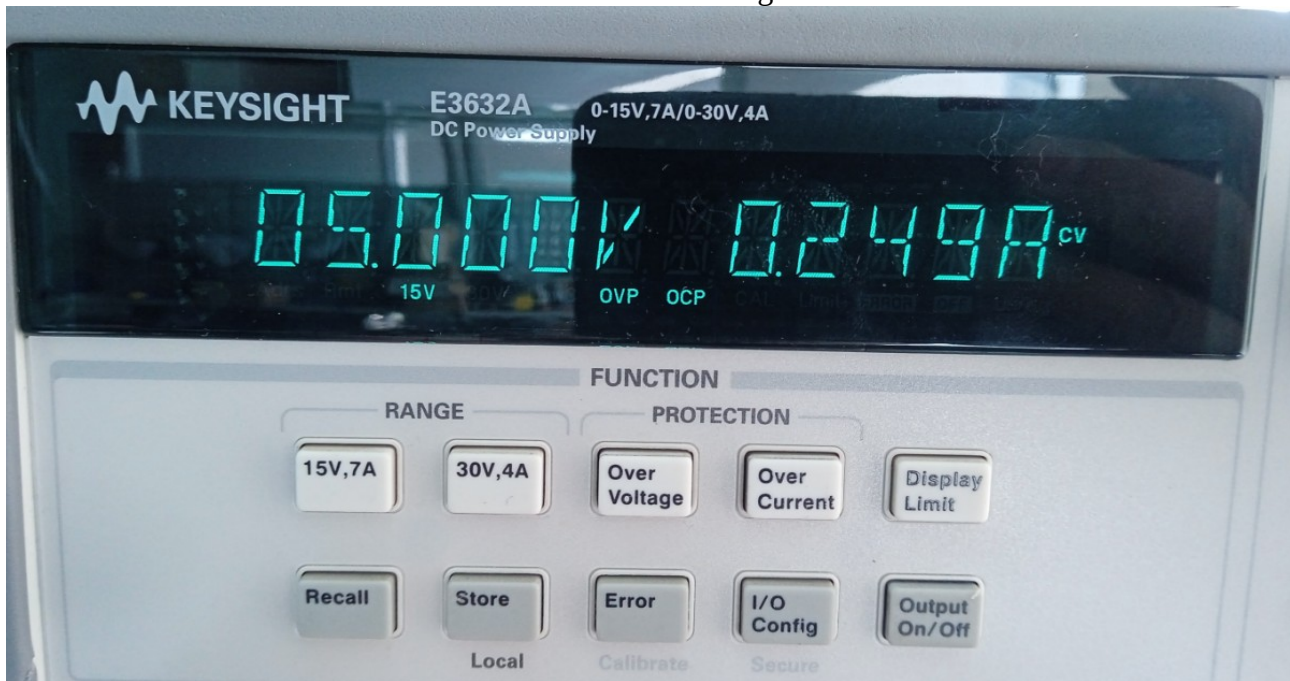
6. PoE Injector Test

Using a straight through cable to connect DPU board to PoE Injector or a network switcher. Poor quality network cable may cause error, the PD can't get power supply from PSE. Copper wires are highly recommended. DPU uses 10/100 BASE-T RJ45 connector, only supports 10/100 BASE-T Power Over Ethernet standard. Both Alternative A and Alternative B are verified successfully. Alternative A uses the center tap of 1-2 pair and 3-6 pair. Alternative B uses the spare wires, 4-5 pair and 7-8 pair. PoE power supply module don't care about the polarity because of a bridge rectifier used.



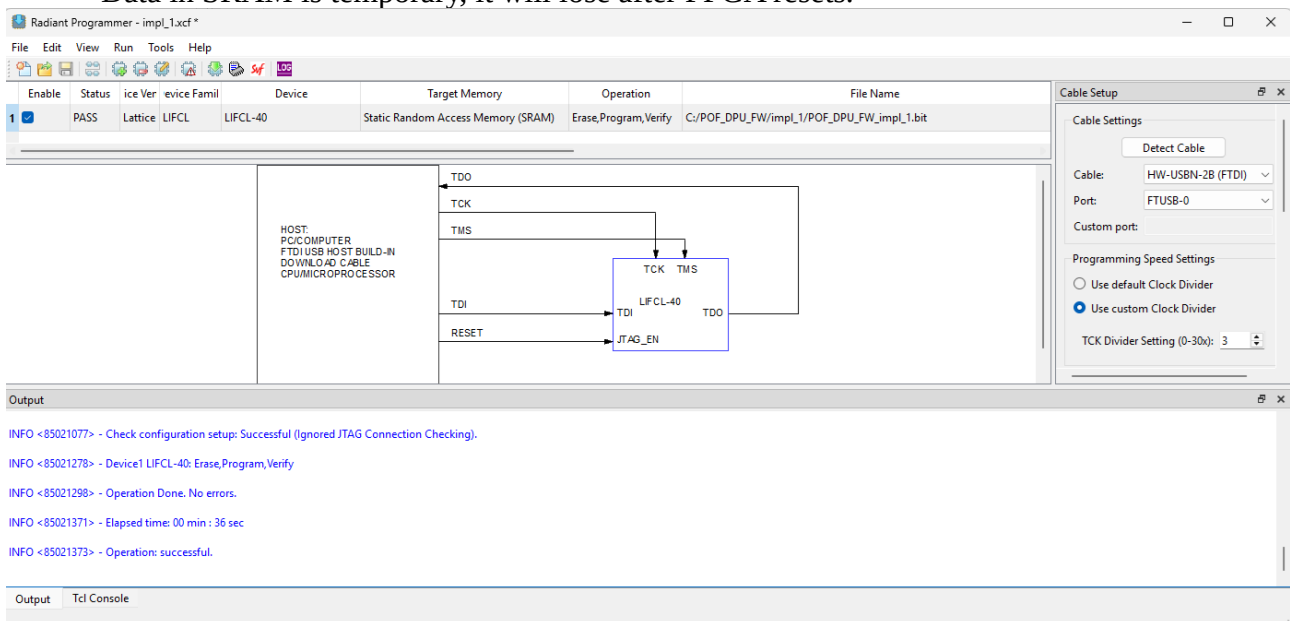
7. Entire Board Power Consumption

Peak current reaches to 0.3A when the infrared image sensor shutter works.



8. SRAM Programming Test

Data in SRAM is temporary, it will lose after FPGA resets.



LIFCL - LIFCL-40 - Device Properties

General

Device Information

Device Operation

Target Memory:

Static Random Access Memory (SRAM) ▾

Port Interface:

JTAG ▾

Access Mode:

Direct Programming ▾

Operation:

Erase, Program, Verify ▾

Programming Options

Programming file:

C:/POF_DPU_FW/impl_1/POF_DPU_FW_impl_1.bit

...

☐ Password Protection Options (Provide key file if password protection enabled)

OK

Cancel

9. SPI Flash Programming Test

LIFCL - LIFCL-40 - Device Properties

General Device Information

Device Operation

Target Memory: External SPI Flash Memory (SPI FLASH) ▾

Port Interface: JTAG2SPI ▾

Access Mode: Direct Programming ▾

Operation: Erase,Program,Verify ▾

Programming Options

Programming file: ./POF_DPU_FW/impl_1/POF_DPU_FW_impl_1.bit ... 0x6206

SPI Flash Options

Family: SPI Serial Flash ▾

Vendor: WinBond ▾

Device: W25Q32JW ▾

Package: 8-pad XSON ▾

SPI Programming

Data file size (Bytes): 811698 Load from File

Start address (Hex): 0x00000000 ▾

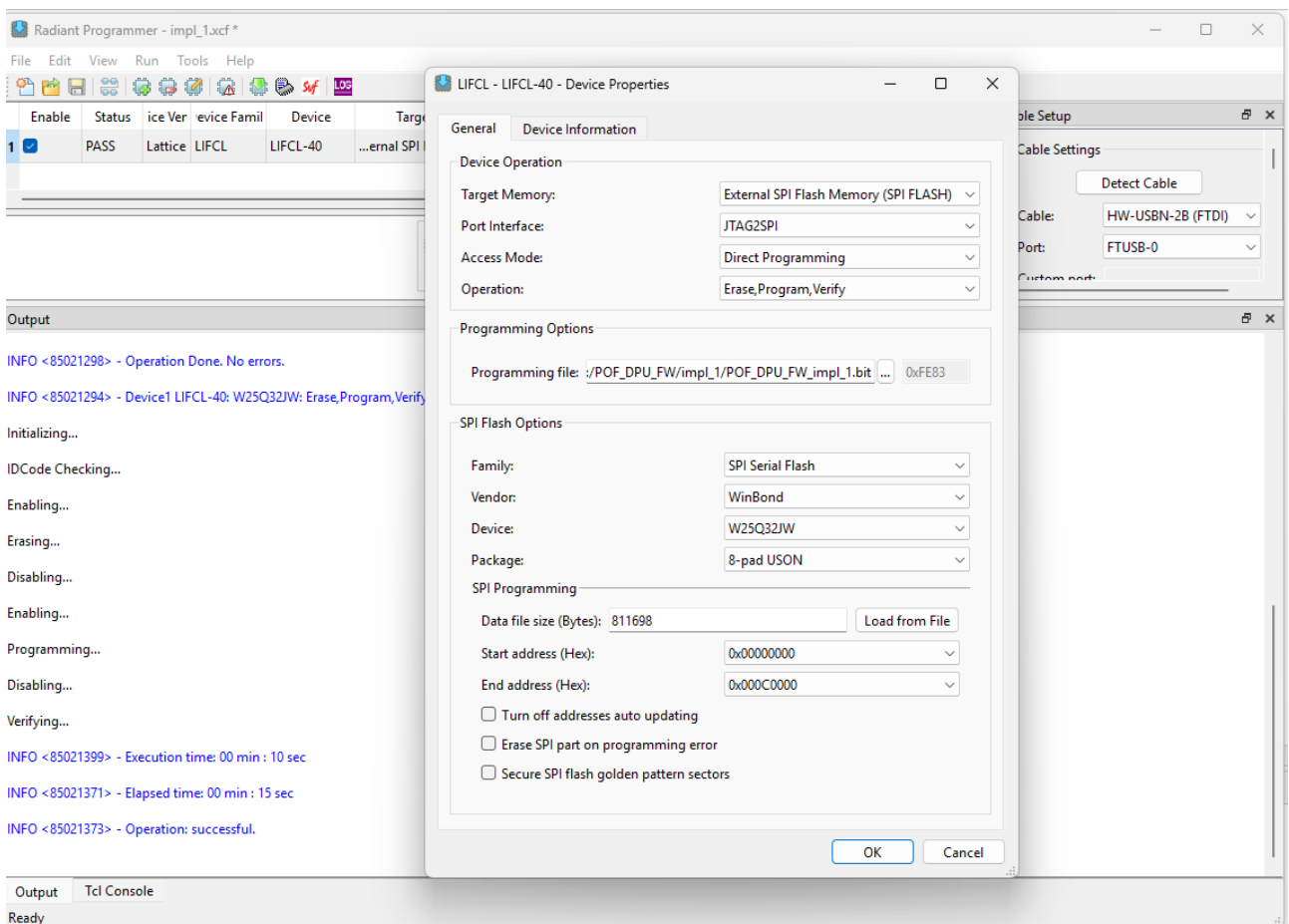
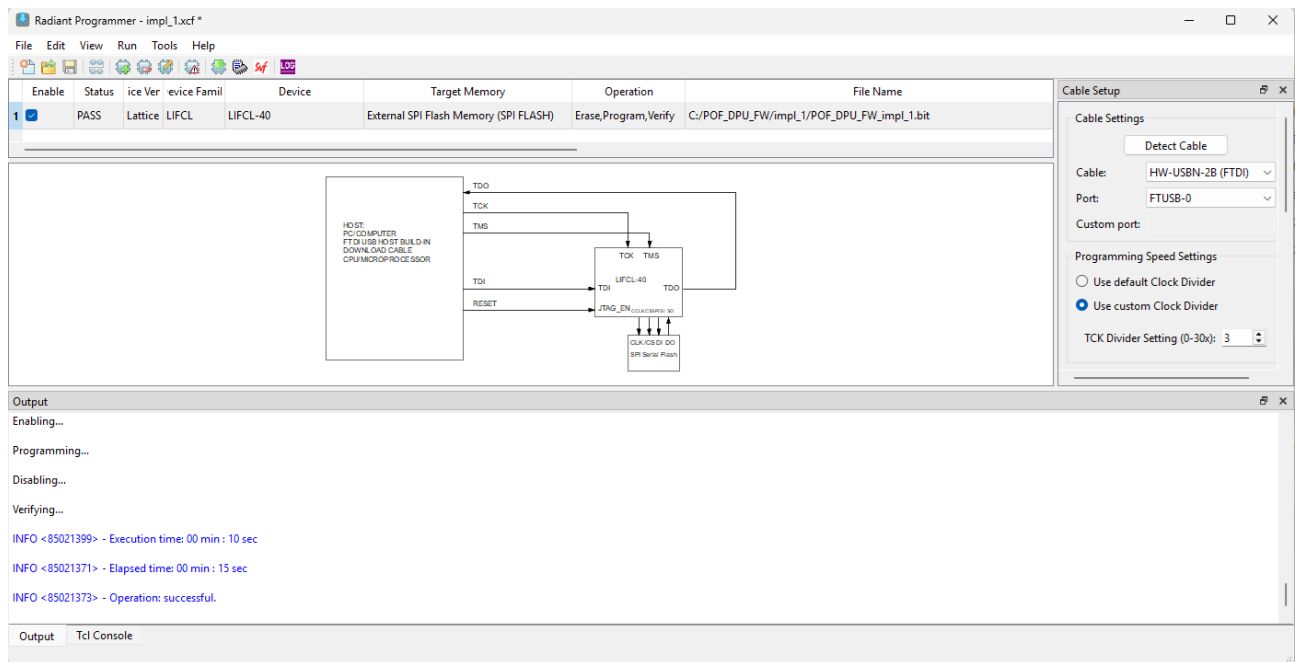
End address (Hex): 0x000C0000 ▾

☐ Turn off addresses auto updating

☐ Erase SPI part on programming error

☐ Secure SPI flash golden pattern sectors

OK Cancel



10. Hyper RAM Read/Write Test

To be continued.

11. OV5640 Visible Light Image Sensor Test

To be continued.

12. InfiRay Lite640 Infrared Image Sensor Test

To be continued.

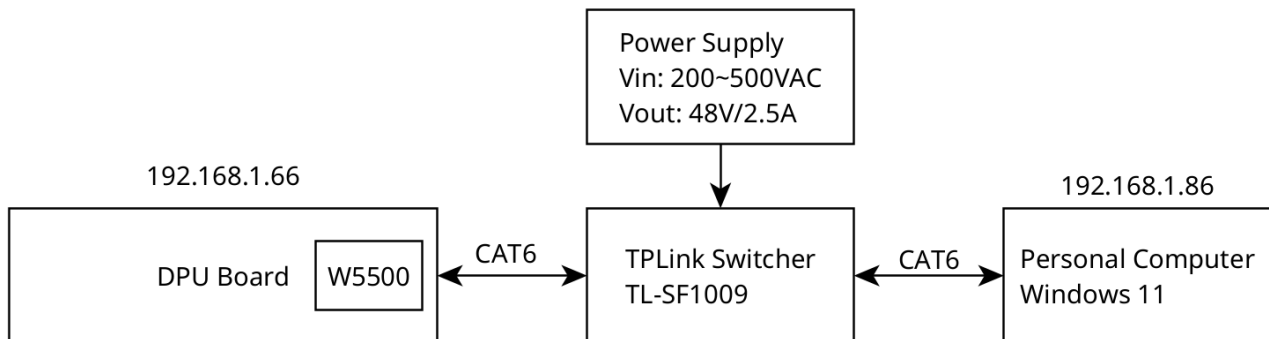
13. AD7988 Converter Test

To be continued.

14. Ethernet W5500 Test

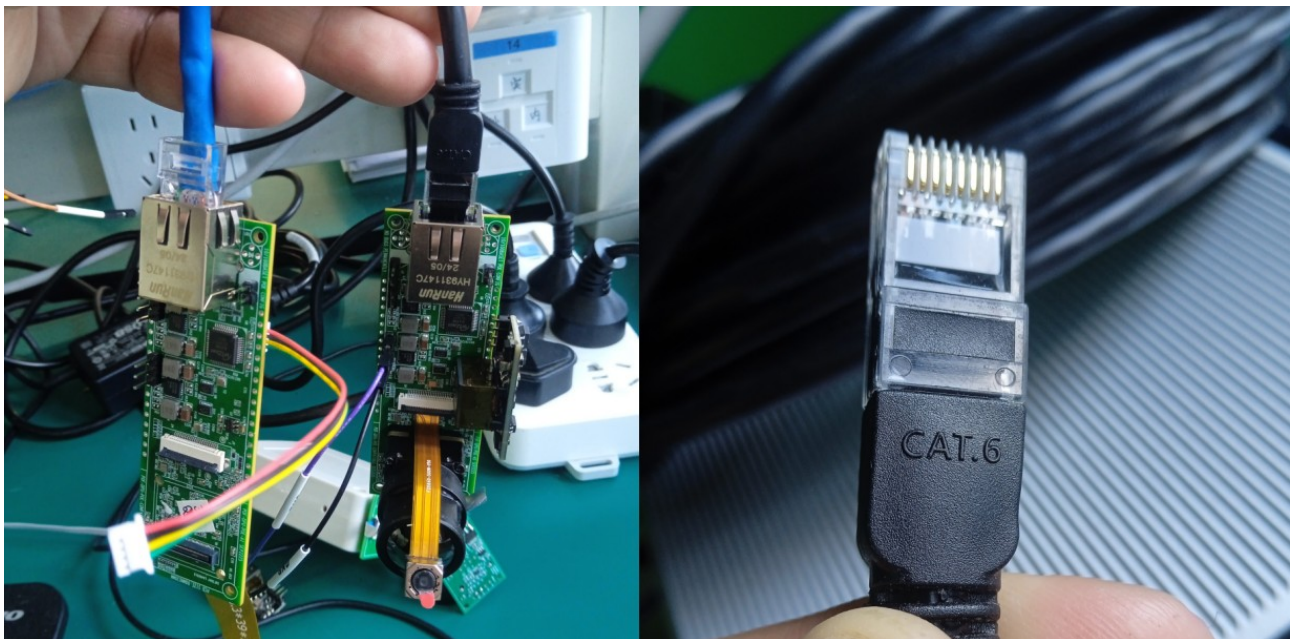
14.1 Network Configuration

We establish a local area network, set the IP address of W5500 to 192.168.1.66, set the IP address of PC to 192.168.1.86. PC works as a TCP server and listens on port 0x1234(4660). Once W5500 starts up, it tries to launch connection until success. It periodically sends fixed pattern “ILoveChina” to server.



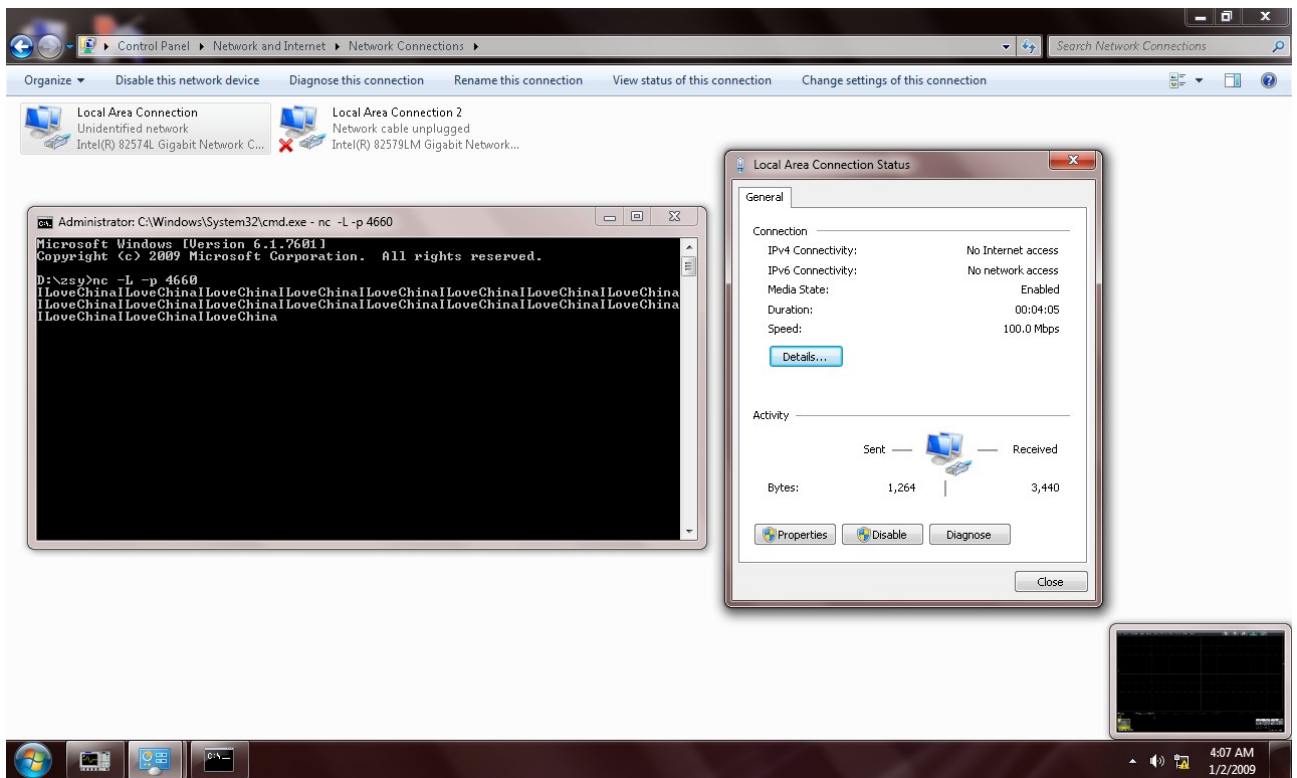
14.2 Hardware Connection

In this case, we use an industrial switcher of TPLink TL-SF1009P which that supports PoE in 8 ports. With zero configuration, connect W5500 and PC to the switcher. Highly recommend to use CAT6 network cable to get better performance.



14.3 Power On and Test

We use nc(NetCat) to listen on port 4660, power on the PoE switcher, after a while, the Auto Negotiation completes, nc receives data from W5500. Warning, in practice, some bad quality network cables may cause ping timeout.



15. Other issues.

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